

The structural evolution of the Chinese aviation network during and after the pandemic: A machine learning-based approach

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ABSTRACT

To identify the evolution pattern of the Chinese aviation network before, during, and after the pandemic, we develop a machine learning-based framework to analyze the network's development dynamics. By integrating link prediction algorithms into this framework, we quantify the contributions of 11 topological features driving structural changes. Utilizing aviation passenger flow data from China from 2014 to 2024, we identify important topological features that reveal the impact of the COVID-19 pandemic on the air network evolution. The empirical findings yield the following insights: (1) Targeted investments in core hub airports should be prioritized, given their critical role in maintaining network connectivity and facilitating rapid recovery during disruptions. (2) Airlines should strategically optimize shared connectivity and resource allocation to maintain critical routes and network resilience during times of resource constraints caused by the pandemic. (3) To control possible cascading effects caused by disruptions on international routes, secondary hubs and regional routes can be promoted. This would stabilize domestic connectivity and enhance the resilience of the aviation network. (4) Post-pandemic, the diversity-driven topological features become more prominent, suggesting airlines' plan of enhancing network robustness. Policymakers should promote the development of secondary hubs and new routes, thereby improving the aviation network's resilience and reducing excessive concentration to core hubs. These findings provide practical insights for balancing centralization, regional development, and network diversification, contributing to a resilient and adaptive aviation network capable of withstanding future disruptions.

1. Introduction

Air transportation is important in enabling travel, trade, and tourism but is highly vulnerable to external shocks such as public health emergencies. The COVID-19 pandemic caused unprecedented disruption to global transportation systems, with the aviation network experiencing probably the most severe impacts (Sun et al., 2021; Zhou et al., 2021a). According to the International Civil Aviation Organization (ICAO), the number of seats on international flights in 2020 was 66% lower than that before the 2019 outbreak, and the number of passengers declined by as much as 74% (ICAO, 2020). This substantial reduction in air travel indicates the vulnerability of the aviation network to global crises. Although the immediate effects of the pandemic on aviation networks have been extensively studied (Sun et al., 2020; Zhang et al., 2025a;

Zhou et al., 2021a), a comprehensive understanding of the structural evolution of the aviation network before, during, and after the pandemic remains limited (Wong et al., 2023). Therefore, it is important to explore how external shocks reshape aviation networks and identify the topological changes that occur during periods of disruption and recovery. This will enable aviation companies to better respond to future disruptions, while providing valuable policy insights for strengthening the resilience and network connectivity of the aviation sector.

Previous studies analyzing the structural evolution of the aviation network have predominantly relied on an experience-driven selection of topological features to describe network structure (Mueller, 2023; Yang et al., 2023; Zhang et al., 2025b). Although this approach is intuitive, it suffers from a critical limitation: the selected features may not accurately capture the true evolutionary characteristics of the network under

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0967-070X/© 2025 Elsevier Ltd. All rights reserved, including those for text and data mining, AI training, and similar technologies.

study, particularly when the network undergoes substantial changes due to external shocks such as the COVID-19 pandemic. Relying on pre-defined metrics overlooks the subtle and dynamic interactions that drive network evolution, limiting the insights that can be drawn into the underlying structural changes. Moreover, traditional methods struggle to evaluate the complex interdependencies among various features, leaving their potential influence on network dynamics unexplored. In recent years, machine learning (ML) has emerged as a powerful tool for analyzing complex and large-scale datasets, offering the ability to uncover hidden relationships and patterns (Lei et al., 2022; Wang et al., 2022). Incorporating multiple features into the ML-based framework enables researchers to identify the most influential features driving network evolution, providing a more comprehensive and accurate representation of structural changes (Xu et al., 2024). However, research has not adequately applied it to identify the determinants of dynamic network evolution in response to major external shocks, such as those caused by the pandemic.

Another limitation of current studies analyzing the evolution of aviation networks during the pandemic stems from the type of data used. Many studies relied on flight schedules (Yang et al., 2023; Zhang et al., 2025a), which deviate significantly from the actual flights performed due to frequent cancellation and rescheduling during the pandemic. In comparison, operational data such as the actual origin-destination (OD) passenger flows provide a more accurate and reliable criterion for network connectivity, because they reflect the actual movement of passengers between origins and destinations. In addition, they reveal travel demand and actual utilization of the aviation network, controlling the very volatile load factors during the pandemic. By analyzing these data, it is possible to identify the real impacts of the COVID-19 pandemic on passenger mobility, route changes, and urban and regional connectivity.

To address these research gaps, this study aims to comprehensively identify the key topological features driving changes in the Chinese civil aviation passenger network and analyze the mechanisms underlying its structural evolution across the pre-pandemic, pandemic, and post-pandemic periods. Specifically, we propose an integrated framework that combines ML and link prediction methods, using actual OD passenger flow data to reveal the key topological features influencing the evolution of the aviation network. The primary contributions of this study are as follows:

- (1) An integrated framework combining ML and link prediction is proposed to capture the evolutionary dynamics of the Chinese aviation network during the COVID-19 pandemic. This approach quantifies the contributions of 11 topological features, overcoming possible bias in subjectively selecting features of network changes.
- (2) By utilizing actual OD passenger flow data, this approach reveals how route changes influenced the network's structure during the pandemic, providing a more accurate assessment of network adaptation and recovery during the pandemic than studies relying on schedule data or aggregated flight data.
- (3) The evolution of China's aviation network across the pre-pandemic, pandemic, and post-pandemic periods is systematically examined, revealing the shifting preferences and priorities in aviation network design. Using SHapley Additive exPlanations (SHAP) analysis, we reveal how the emphasis on centralization, resource optimization, and regional connectivity changes under changing conditions.

The rest of this paper is organized as follows. Section 2 reviews the relevant literature. Section 3 introduces the ML-based framework developed to predict the evolution of the aviation network during crises such as the pandemic. In Section 4, we use real-world data to train and test the model, generating prediction results and identifying the key features that drive the prediction process. Section 5 discusses the policy

implications. Finally, Section 6 summarizes the main conclusions.

2. Literature review

This section reviews two streams of the literature, focusing on (1) the pandemic's impact on the structure of the aviation network and (2) link prediction in aviation networks. In the next section, we highlight our contributions by comparing our work to these studies.

2.1. The pandemic's impact on the structure of the aviation network

The COVID-19 pandemic has prompted research into its impact on the structural evolution of aviation networks (Li et al., 2022). Most studies have used network science and graph theory to analyze the structural evolution of aviation networks. Table 1 compares studies analyzing the evolution of aviation networks during the pandemic, highlighting their differences in methodologies, feature identification approaches, and datasets. Metrics such as degree centrality, inter-centrality, and network density have been used to assess changes in connectivity and hub dominance. Sun et al. (2020) investigated the impacts of COVID-19 on global air transport and the dynamics of spatiotemporal evolution at different scales, using airports and countries as network nodes, respectively. Zhou et al. (2021b) proposed weighted international efficiency metrics and international robustness metrics to measure the ability of a country's aviation network to maintain international connectivity and robustness, respectively, during the pandemic. Li et al. (2021) used network connectivity and airport centrality metrics to analyze changes in aviation networks during the pandemic. Zhou et al. (2021a) proposed a layered weighted network efficiency metric to measure air transportation network connectivity and study its vulnerability to different levels of disruption. The results identified the most critical airport connections and their impact on network connectivity. Cheung et al. (2022) recommended that airlines establish strategic partnerships with other carriers to enhance transfer services and improve connectivity. Through a quantitative analysis, Zhang et al., 2025a revealed that the COVID-19 pandemic led to a major reorganization and decentralization of the aviation network structure from the traditional core-periphery model to a hybrid structure of core-periphery mixed with local communities. However, conventional network analysis methods struggle to clarify the complex interdependencies between network features, limiting their ability to identify the key topological factors. Most prior studies rely on a limited number of manually selected indicators and do not provide quantitative assessments of feature importance. For example, Li et al. (2021) and Zhou et al. (2021a) use classical metrics such as degree or betweenness centrality, but they lack a systematic framework for evaluating their relative contributions. In contrast, we integrate ML-based classification with link prediction models to assess the combined effects of 11 similarity-based features, selected through empirical training.

Table 1
Research on the evolution of aviation networks during the pandemic.

Literature	Identification of important features	Multiple topological features	Dataset
Sun et al. (2020)		✓	Flight schedule data
Zhou et al. (2021b)		✓	Flight schedule data
Li et al. (2021)			Flight schedule data
Zhou et al. (2021a)			Flight schedule data
Zhang et al. (2025a)			Flight schedule data
This study	✓	✓	Passenger flow data

Furthermore, the use of SHAP allows us to clearly track how the importance of each indicator changes over different stages of network evolution, a perspective that is often missing in previous studies.

Moreover, many studies on the evolution of aviation networks have relied solely on flight schedule data (Li et al., 2021a; Zhou et al., 2021b). However, flight schedules are often subject to operational disruptions such as cancellations and delays. These factors are especially prevalent during the COVID-19 pandemic and can lead to substantial differences between scheduled services and actual passenger movements. To better capture real network evolution, we use OD passenger flow data, which provide more accurate information on how passengers travel between airports. Additionally, several studies have examined the impact of COVID-19 on aviation networks, but most of them focus only on the disruption period and do not include a comparison of the recovery process. In contrast, our study covers the entire timeline, including the pre-pandemic, pandemic, and post-pandemic periods. This allows us to analyze how different network priorities, such as centralization, resource efficiency, and regional connectivity, changed at different stages of the crisis.

2.2. Link prediction in aviation networks

Aviation stakeholders (e.g., airlines and airports) constantly optimize their own aviation networks and, meanwhile, attempt to predict the changes of the whole system (Bayram et al., 2023). Link prediction algorithms, known for their computational simplicity and strong predictive performance, have been widely used to identify potential connections between two nodes within a network (Lü and Zhou, 2011). Table 2 provides a comprehensive overview of advances in link prediction methods.

Traditional link prediction methods for network evolution have often relied on predefined or commonly used features to guide the prediction process (Kumar et al., 2020; Mueller, 2023). For example, Zhang et al. (2024) applied the link prediction method to develop an indicator system that incorporates both endogenous network factors and external attributes and further analyze the predictive performance of each indicator. Although this approach is straightforward, it suffers from an important limitation: the selected features may fail to accurately reflect the true evolutionary characteristics of the network under study. For instance, in centralized networks, using the “hub depressed” (HD) feature may fail to accurately reflect the network’s dominant features, leading to suboptimal or even misleading predictions. Therefore, it is important to select the features that are most relevant to the dynamics of

Table 2

Current research on the analysis of network evolution using the link prediction method.

Literature	Rapidly changing networks	Dataset	Decisions of link prediction		
			Add	Retain	Remove
Bliss et al. (2014)	✓	Daily, Panel	✓		
Fu et al. (2018)		Cross-Sectional	✓		✓
Zhang et al. (2018)		Cross-Sectional	✓		
Zhang et al. (2020)		Cross-Sectional	✓		
Lei et al. (2022)	✓	Monthly, Panel		✓	✓
Mueller (2023)		Annual, Panel	✓		
B. Zhang et al. (2024)		Annual, Panel	✓		
P. Zhang et al. (2024)	✓	Annual, Panel	✓		
This study	✓	Monthly, Panel	✓	✓	

Table 3

Network definitions.

Measure	Symbol	Description
Node	N	An airport within the Chinese network
Node degree	K	The total number of connections node i has with other nodes
Link	L	An air route within the Chinese network
Weight	W	The amount of passenger traffic between nodes i and j
Distance	D	The distance between airports i and j in link l

the network’s evolution. Conversely, these studies do not provide a clear framework for identifying which topological features drive such changes or how their importance evolves over time.

Efficient computational methods are needed for the prediction of dynamic changes in a network, if there are a large number of nodes and rapidly changing edges (i.e., air routes). ML has important advantages in handling link prediction in complex large-scale networks (Lei et al., 2022). ML models predict potential future links by learning from historical link data, using feature vectors that incorporate topological features and node attributes. Sophisticated classification algorithms, such as random forest (Breiman, 2001) and XGBoost (Chen and Guestrin, 2016), have been widely used to classify and predict future network behavior. Mueller (2023) combined a similarity-based link prediction approach with a supervised recurrent neural network framework to predict links in a European aviation network. Lei et al. (2022) used ML classification algorithms to predict edge removal decisions in Brazilian public transport and US aviation networks. However, these studies primarily focus on stable network structures and are not fully applicable to emerging or rapidly evolving systems.

To address these research gaps, our study focuses on a highly dynamic network, where link additions and removals occur frequently due to market changes and external shocks. We propose an ML-based link prediction framework that not only incorporates 11 similarity-based topological features but also uses SHAP to quantify the relative importance of each feature across different time periods. This allows us to move beyond black-box prediction toward interpretable and adaptive modeling, which is essential for real-world decision-making of air network design in uncertain contexts.

3. Methodology

3.1. Transportation network

In this study, we construct an undirected temporal network G_t , where t indexes the month under consideration. Table 3 shows the key measures and definitions in the Chinese network analysis. The aviation network is represented by a graph $G_t(V_N, E_L)$. Here, V_N is the set of N airport nodes and E_L comprises a set of L air routes connecting these nodes. In the Chinese aviation network, each node corresponds to a city hosting an airport. Each link in this network represents at least one operational route connecting two cities that is serviced by any airline during a given month. We assume that the network is undirected, and an analysis of data from Official Airline Guide (OAG) supports this hypothesis, because the passenger flows between almost all pairs of nodes are quite equal in both directions.

Furthermore, weight represents the importance and utilization of each route, providing key insights into the characteristics of network dynamics. We thus construct a weighted aviation network where the weight of each link is defined as the average passenger flow between two nodes i and j ($i, j \in N$). We aggregate nodes, edges, and weights on a monthly basis, creating network snapshots over a 10-year period. These snapshots capture the monthly dynamics of the Chinese aviation network, such as changes in connectivity and traffic flow, offering a detailed view of its temporal evolution.

3.2. ML-based prediction framework

Fig. 1 shows the research framework of the ML-based prediction model for the aviation network. The proposed framework includes feature selection, modeling, evaluation, and interpretation. The prediction process commences with the definition of the Chinese aviation network. To accurately predict changes in the network, the model uses a similarity-based algorithm of the link prediction method by calculating the value of 11 topological features (as detailed in Section 3.3). Table A.6 in Appendix A1 summarizes the input variables used in the ML-based model. The pseudocode of the ML-based algorithm is presented in Appendix A2.

The link prediction problem is often considered a binary classification task (Zhang et al., 2018), i.e., the model needs to decide whether each edge in the network should be added. We consider all nodes in the network—instead of a predefined subset of nodes—as potential candidates for establishing air route connections. This increases the complexity of the problem (Silva and Cunha, 2017), but nevertheless avoids possible bias introduced in subset choices. Two models are proposed. The first model is an unweighted topological feature model, which is used to examine the network structure without considering the weight of the links. The other model is a weighted topological feature

model that uses the passenger volume on each link as a weight, resulting in a more detailed and accurate description of the network dynamics. These models are trained and tested using a 10-year dataset of the Chinese aviation network, with the data divided into a 70% training set and a 30% testing set to ensure robustness. A 10-fold cross-validation method is used to validate the performance of the model, and both simultaneous and non-simultaneous testing methods are used to assess the adaptability and accuracy. Specifically, the simultaneous test is performed on the same graph snapshot used for training, evaluating the immediate predictive accuracy. The non-simultaneous test involves testing on subsequent graph snapshots to assess the model's ability to generalize and predict future network changes based on historical dynamics.

Next, for model evaluation, metrics such as the confusion matrix, precision, and accuracy are used to assess the model's predictive performance (as detailed in Section 3.4).

Finally, SHAP analysis is used to interpret the model and identify the key topological features driving the evolution of the network. This integrated framework effectively combines ML with network analysis, providing a robust approach to understanding and predicting the dynamic evolution of the aviation network.

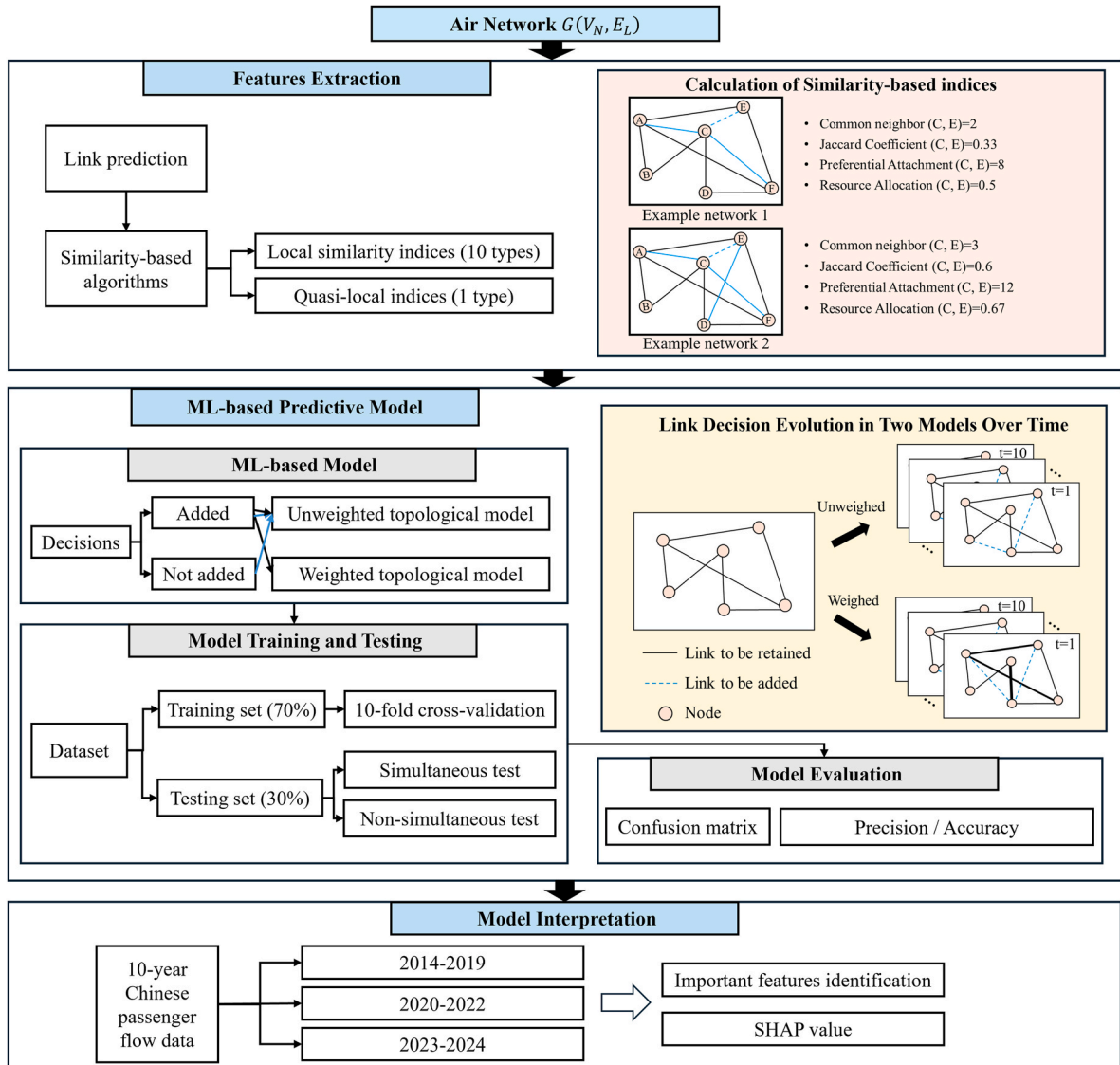


Fig. 1. Structure of the machine learning-based network prediction process.

3.3. Feature metrics

We consider a subset of possible unweighted topological features used widely in link prediction literature. Among these, 10 features are local similarity indices (Lei et al., 2022), and one is a quasi-local index. Local indices are calculated by using information about common neighbors and node degree (Kumar et al., 2020). We also introduce quasi-local indices as a trade-off between the local and global approaches, or between performance and complexity.

- (1) Common neighbor (CN) index: This metric quantifies the direct overlap in the neighborhood of nodes m and n , measured by counting the number of nodes that are adjacent to both.

$$|\Gamma_m \cap \Gamma_n| \quad (1)$$

- (2) Salton (SA) index: This measure evaluates the proportion of shared neighbors between nodes m and n , adjusted by the geometric mean of their degrees to normalize the effect of node connectivity.

$$\frac{|\Gamma_m \cap \Gamma_n|}{\sqrt{k_m \times k_n}} \quad (2)$$

- (3) Jaccard (JA) index: The number of shared neighbors divided by the total number of distinct neighbors between the two nodes.

$$\frac{|\Gamma_m \cap \Gamma_n|}{|\Gamma_m \cup \Gamma_n|} \quad (3)$$

- (4) Sorensen (SO) index: This index normalizes the shared neighbors by the average degree of the two nodes, assigning higher similarity to node pairs with lower overall connectivity.

$$\frac{2|\Gamma_m \cap \Gamma_n|}{k_m + k_n} \quad (4)$$

- (5) Hub promoted (HP) index: This index amplifies similarity when at least one of the nodes has a relatively low degree, using the minimum degree as the normalization factor.

$$\frac{|\Gamma_m \cap \Gamma_n|}{\min(k_m, k_n)} \quad (5)$$

- (6) Hub depressed (HD) index: In contrast to HP, this index penalizes similarity when one node is highly connected, normalizing by the higher degree of the two.

$$\frac{|\Gamma_m \cap \Gamma_n|}{\max(k_m, k_n)} \quad (6)$$

- (7) Leicht Holme Newman (LHN) index: The normalized count of common neighbors, obtained by dividing the number of shared adjacent nodes by the product of the degrees of the two nodes.

$$\frac{|\Gamma_m \cap \Gamma_n|}{k_m \times k_n} \quad (7)$$

- (8) Preferential attachment (PA) index: Based solely on node degrees, this index assumes that high-degree nodes are more likely to form links and is computed as the product of their degrees.

$$k_m \times k_n \quad (8)$$

- (9) Adamic adar (AA) index: This index weighs common neighbors inversely by the logarithm of their degree, assigning more importance to less-connected shared neighbors.

$$\sum_{z \in \Gamma_m \cap \Gamma_n} \frac{1}{\log k_z} \quad (9)$$

- (10) Resource allocation (RA) index: This index weighs common neighbors inversely by the logarithm of their degree, assigning more importance to less-connected shared neighbors.

$$\sum_{z \in \Gamma_m \cap \Gamma_n} \frac{1}{k_z} \quad (10)$$

- (11) Local path (LP) index: The first term in Equation (11) is the number of paths of length equal to 2 between the nodes m and n , and the second term is the number of paths of length equal to 3 between the nodes m and n damped by parameter ξ .

$$S_{mn,2} + \xi \cdot S_{mn,3} \quad (11)$$

3.4. Model evaluation and interpretation

This section outlines the evaluation approaches used to comprehensively assess prediction performance. First, the confusion matrix, an effective tool for evaluating the results of link prediction (Kumar et al., 2020), records four scenarios of model classification: true positive (TP), false positive (FP), false negative (FN), and true negative (TN). Based on these values, we compute precision and accuracy, respectively. Precision represents the proportion of true cases in all samples with positive predictions (Herlocker et al., 2004). Accuracy measures the proportion of correctly predicted instances. The equations for calculating these metrics are as follows:

$$\text{Precision} = \frac{TP}{TP + FP} \quad (12)$$

$$\text{Accuracy} = \frac{TP + TN}{TP + FP + FN + TN} \quad (13)$$

Moreover, we use SHAP value to quantify the contribution of each topological feature to the predictive outcomes (Spadon et al., 2019). By ranking feature importance across all possible inclusion orders, SHAP value ensures an unbiased assessment of each feature's true impact on the model's predictions. This approach provides a clear interpretation of how key network characteristics influence aviation network evolution.

Using Chinese aviation data spanning 10 years, we use the Kolmogorov–Smirnov test to conduct a preliminary comparison of the feature samples of retained versus added edges. Because multiple comparisons are involved, we apply Bonferroni corrections to maintain a stringent significance level. We further refine our feature selection by performing stratified 10-fold cross-validation on a balanced training dataset, using five classification algorithms from the scikit-learn library (i.e., RandomForestClassifier, XGBClassifier, GaussianProcessClassifier, DecisionTreeClassifier, and GradientBoostingClassifier). Table 4 presents the

Table 4
Comparison of prediction results.

Algorithm	Unweighted model		Weighted model	
	Precision	Accuracy	Precision	Accuracy
Decision Tree	0.774	0.740	0.781	0.760
Random Forest	0.781	0.765	0.795	0.782
Gaussian Process Classification	0.722	0.690	0.735	0.701
Gradient Boosting Decision Trees	0.802	0.797	0.818	0.830
XGBoost (this study)	0.827	0.814	0.844	0.841

precision and accuracy of the models applied to the Chinese aviation network. We select the XGBClassifier based on its highest performance with the lowest error variance, which are typical of network structures (Lei et al., 2022).

4. Case study

4.1. Data description

In this study, we compile a comprehensive dataset comprising all domestic passenger air routes within mainland China for the period of June 1, 2014 to June 1, 2024. The data are derived from the Cirium database, a global aviation data provider, and represent monthly OD passenger flows. The dataset used in this study is derived from the passenger aircraft network and reflects operated flights by civil passenger carriers. While our analysis focuses on passenger transport, the data also indirectly reflect air cargo activity, since a significant portion of air cargo in China is transported via the belly space of passenger aircraft (Deng et al., 2022; Gong et al., 2018). Therefore, although we do not analyze dedicated freighter operations, the network structure captured here is representative of both passenger connectivity and a large share of domestic air cargo flow. Moreover, unlike many existing studies that rely on scheduled flight data, our dataset is based on actual passenger flow records, which more accurately reflect real travel behavior. This is important during the COVID-19 pandemic, when scheduled services often diverged significantly from realized traffic.

Our dataset provides a detailed monthly snapshot of the air travel landscape over a decade and comprises 1,582,280 OD-level records. This dataset allows us to track route-level dynamics and detect subtle shifts in network connectivity and passenger demand. For example, the Chinese aviation network contained 249 nodes and 2106 links as of June 2023. The key variables used in the initial dataset are summarized in Table 5. The long time span and high resolution of the dataset support a more robust analysis of network evolution, improving the accuracy and generalizability of the results compared to studies using shorter or scheduled data.

To ensure the reliability of the analysis, we preprocess the raw data. This step includes removing outliers and records with unusually low ridership that do not reflect typical operational patterns. Specifically, the minimum presence of a link between two nodes is defined as 50 flights per year; links with fewer than 50 flights per year are deleted, as they probably were wrongly captured charter or aircraft ferry/technical services. Additionally, the dataset of each month is divided into a training set and a testing set, which respectively make up 70% and 30% of the dataset.

Based on the cleaned dataset, Fig. 2(a) illustrates the number of monthly passengers in the Chinese aviation network from 2014 to 2024. Between 2014 and 2019, passenger traffic grew steadily, reflecting the

consistent expansion of the aviation industry. However, the COVID-19 pandemic in early 2020 caused a sharp decline in passenger volume, which reached its lowest point in the first half of the year. The recovery period began toward the end of 2020, but frequent fluctuations occurred due to successive pandemic waves. By 2023, passenger numbers had stabilized, and in 2024, they approached pre-pandemic levels. Fig. 2(b) presents the monthly fraction of air routes removed from the network during the same period. From 2014 to 2019, the proportion of removed routes remained relatively stable, with minor fluctuations attributable to seasonal adjustments or operational optimizations. In 2020, the pandemic caused a sharp spike in route removals due to collapsing demand (Zhou et al., 2021a; Czerny et al., 2021). By 2024, the removal rate stabilized at slightly higher than pre-pandemic levels, reflecting the structural adjustments made during the crisis. The alignment of peaks and troughs in Fig. 2(a) and (b) underscores the strong relationship between passenger demand and route adjustments.

4.2. Prediction results

The monthly averages of the confusion matrices of the unweighted and weighted models across the pre-pandemic, pandemic, and post-pandemic phases are calculated, as shown in Figs. 3–5. A comparison of these figures shows that both weighted and unweighted models demonstrate a relatively high prediction accuracy, with the weighted model performing slightly better. For instance, in Fig. 3(a), the simultaneous test results for the weighted model show a prediction accuracy of 85% for the “Retained” category and 90% for the “Added” category. In contrast, the prediction accuracy of the unweighted model drops slightly, to 87%, for the “Added” category, suggesting that incorporating weights (such as passenger flow) enhances the model’s predictive accuracy. The overall prediction accuracy decreases slightly due to the shocks associated with the pandemic period, albeit at a satisfactory level. A similar trend is observed in the non-simultaneous test. Specifically, Fig. 4(b) shows that the unweighted model achieves an accuracy of 74% for the “Retained” category and 77% for the “Added” category, while the weighted model improves these accuracies to 76% and 78%, respectively.

4.3. Model interpretation

Figs. 6–8 illustrate the SHAP values of various topological features of the ML model across three distinct periods: pre-pandemic (2014–2019), pandemic (2020–2022), and post-pandemic (2023–2024). Each dot on the plot represents the SHAP value of a specific feature for an individual prediction. The SHAP value distribution graph uses a color gradient from blue to red to visually represent the magnitude and direction of the impact. Specifically, blue dots correspond to lower SHAP values, indicating a negative impact on the model output, which is associated with predictions of edge retention or removal. In contrast, red dots indicate higher SHAP values, representing a positive impact linked to edge addition. The following section analyzes the role of SHAP values in shaping the structural evolution of the aviation network from 2014 to 2024.

(1) The Pre-Pandemic Period (2014–2019):

Fig. 6 reveals that the top eight features influencing the aviation network, as identified by SHAP values, are consistent across both the weighted and unweighted models in the pre-pandemic period. The Preferential Attachment (PA) measures the product of the degrees of the two nodes (Kumar et al., 2020), and therefore the feature reflects the tendency of nodes within a network to connect to highly connected nodes. The PA feature effectively captures this tendency in the Chinese aviation network, where new air routes are preferentially allocated to well-established hub airports with high connectivity. From 2014 to 2019, the impact of PA and Hub Promoted (HP) features is positive in

Table 5
Definitions of the dataset.

Content	Type	Description	Data example
Year	Numerical	Year of the flight (2014–2024)	2023
Month	Numerical	Month of the flight	1
Aircraft type	Categorical	Type of aircraft	Boeing 737
Operating carrier	Categorical	Airline of the flight	MU
Origin	Numerical	Origin airport of the air route	PEK
Destination	Numerical	Destination airport of the air route	PVG
Frequent	Numerical	Frequency of each airline on each route	46 (times/month)
Seats	Numerical	Capacity of each flight	245 (seats)
Passenger number	Numerical	The total number of passengers on each flight	147 (person-trips)
Distance	Numerical	Actual distance traveled along each route	682 (km)

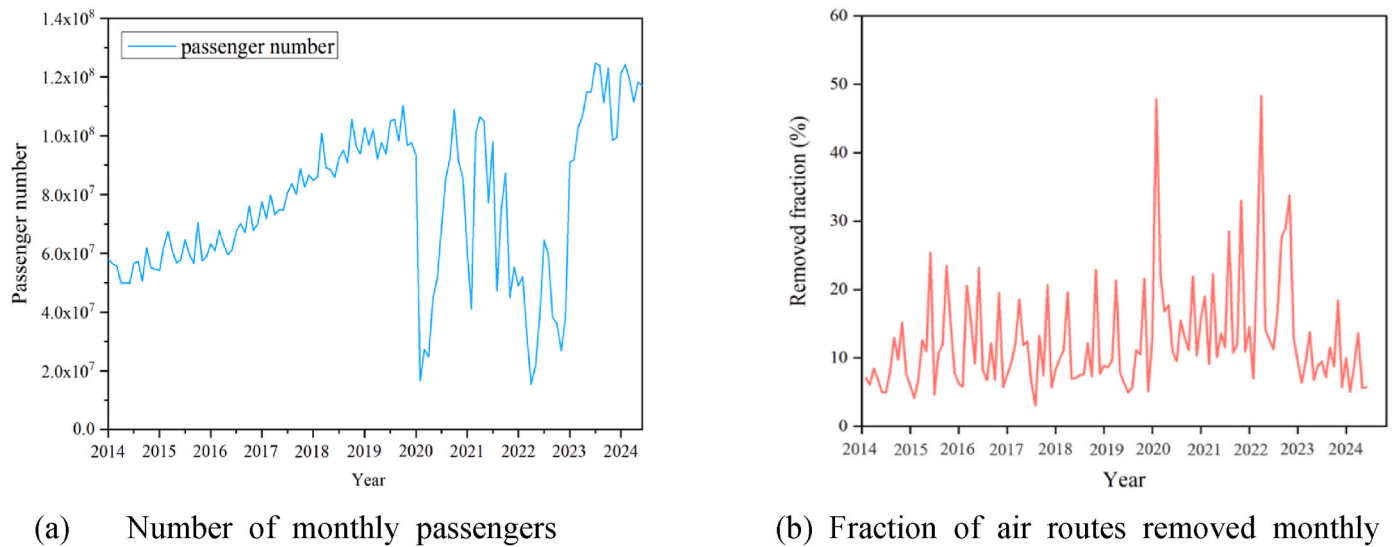


Fig. 2. (a) Number of monthly passengers (b) Fraction of air routes removed monthlyChanges in the Chinese aviation network from 2014 to 2024.

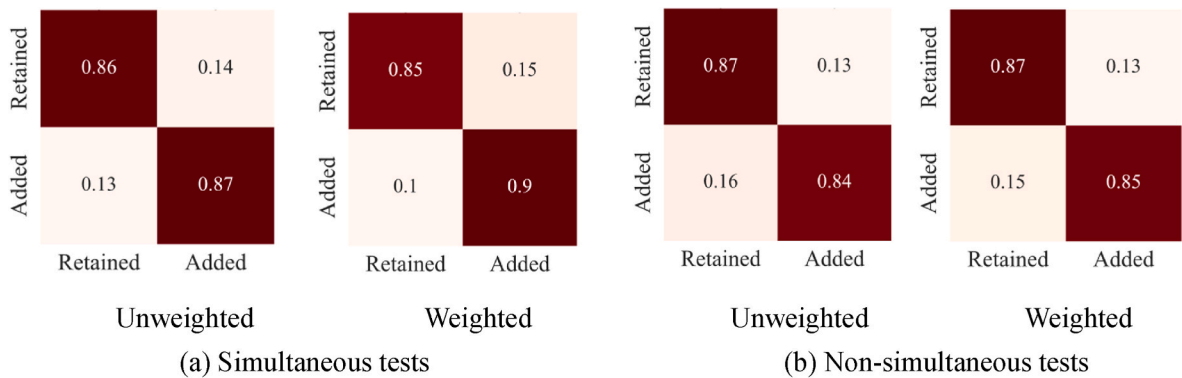


Fig. 3. (a) Simultaneous tests (b) Non-simultaneous testsConfusion matrix of simultaneous and non-simultaneous tests between 2014 and 2019.

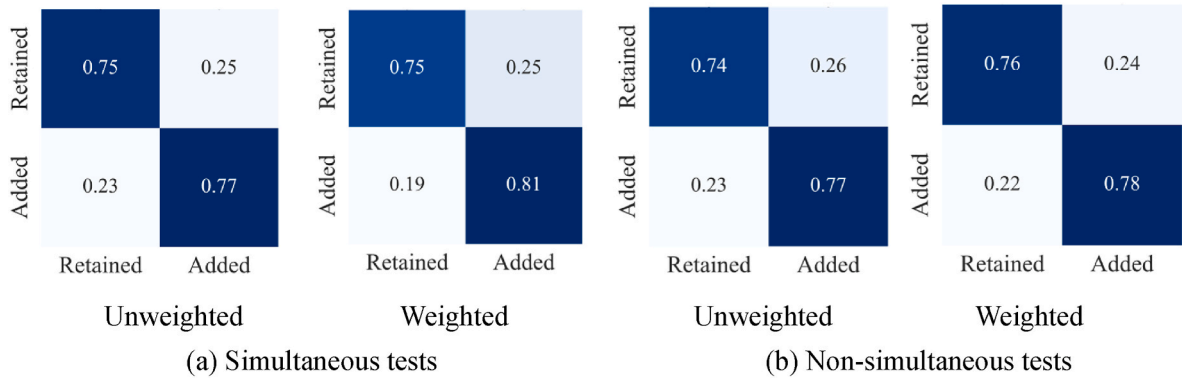


Fig. 4. (a) Simultaneous tests (b) Non-simultaneous testsConfusion matrix of simultaneous and non-simultaneous tests between 2020 and 2022.

both models. The higher value indicates that a hub airport is well-connected to other major nodes, reinforcing its central role in network connectivity and capacity allocation (Alumur et al., 2021). However, while hub airports enhance network connectivity, over-centralization introduces vulnerabilities. A high reliance on a few central hubs increases the risk of cascading failures, where disruptions can significantly reduce network stability and resilience.

The Resource Allocation (RA) feature reflects the distribution of resources from hub airports to their connected airports, emphasizing the

role of shared neighbors and their importance in the resource allocation process (Mueller, 2023). Fig. 6 shows that the RA feature has a particularly strong positive SHAP impact in the weighted model, indicating that the aviation network prioritized optimizing core node connectivity through resource centralization before the pandemic. However, such resource concentration risks marginalizing smaller airports, leading to the contraction of routes serving surrounding airports and weakening the balance of regional networks.

These results indicate that in the pre-pandemic period, airlines

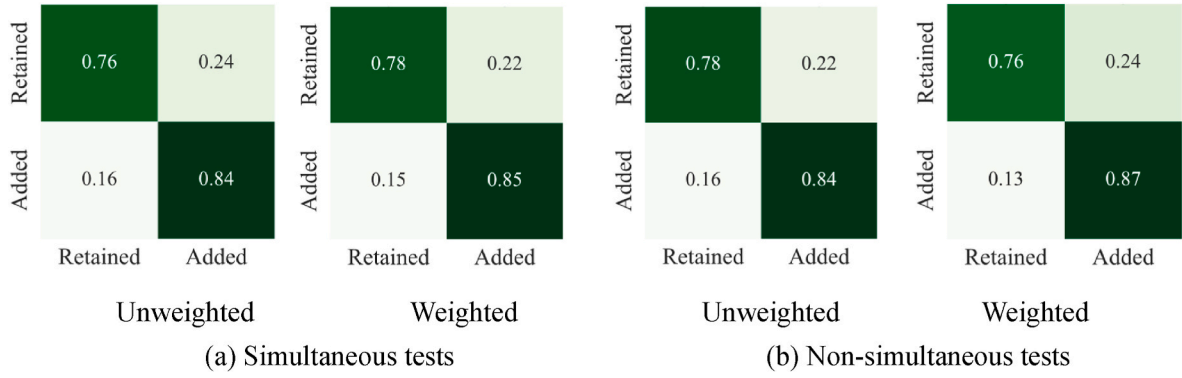


Fig. 5. (a) Simultaneous tests (b) Simultaneous testsConfusion matrix of simultaneous and non-simultaneous tests between 2023 and 2024.

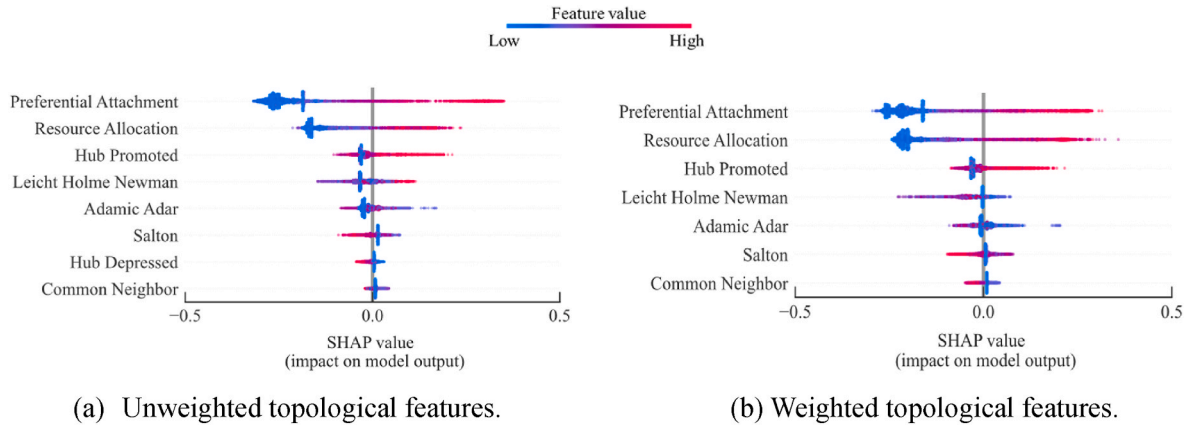


Fig. 6. (a) Unweighted topological features (b) Weighted topological features SHAP values of two models between 2014 and 2019.

prioritized hub airports as central nodes to achieve greater efficiency through traffic concentration and resource optimization.

(2) The Pandemic Period (2020–2022):

From 2020 to 2022, the COVID-19 pandemic substantially altered the topological characteristics of the aviation network and reshaped their relative importance. The SHAP values in Fig. 7 reveal that the network evolved from an efficiency-centered structure to a more decentralized and regionalized configuration, driven by reduced passenger flows and increased operational uncertainties. In both the weighted and unweighted models, the PA feature consistently ranks as the top positive feature influencing route addition decisions,

underscoring the dominance of core hub airports in sustaining connectivity during the pandemic. However, in the weighted model, the diminished importance of the PA feature reflects the substantial decline in passenger traffic. This decline corresponds to strategic route adjustments, particularly the restriction of routes connected to pandemic epicenters, which proved effective in curbing the spread of the pandemic (Lu et al., 2021).

The Leicht Holme Newman (LHN) feature measures the relative connectivity between two nodes while considering the differences in their degree distribution. It captures the tendency for new routes to form between airports with similar connectivity patterns and network attributes, highlighting the interdependence between hubs. During the pandemic, regional airports and smaller-scale connections played a

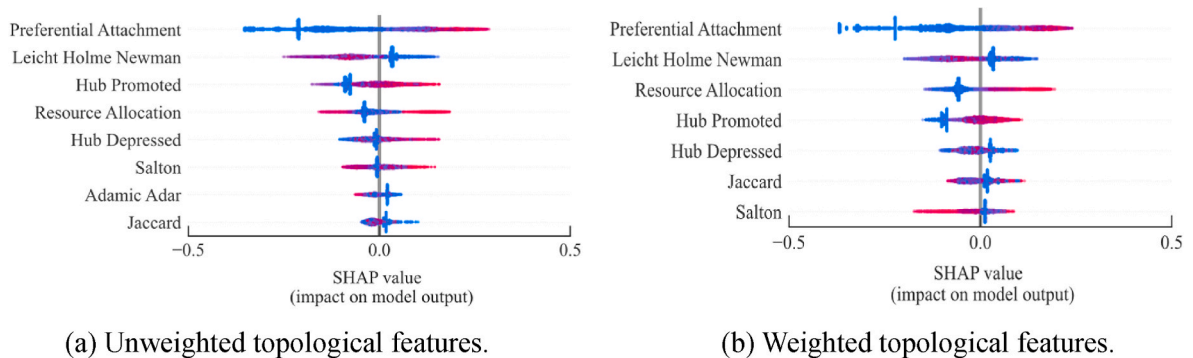


Fig. 7. (a) Unweighted topological features (b) Weighted topological features SHAP values of two models between 2020 and 2022.

critical role in maintaining network connectivity, as major hubs experienced significant traffic reductions due to travel restrictions. A high LHN ranking suggests that many of these smaller airports formed localized clusters, reinforcing regional connectivity even when long-haul routes were severely disrupted. Similarly, the HD feature becomes much more important in the unweighted model, indicating that the pandemic substantially weakened the connectivity of core hub airports. In the weighted model, the HD feature has a pronounced negative impact, reflecting the severe reduction in passenger flow through major hubs during the pandemic. The AA feature highlights the importance of the existing critical routes. In the unweighted model, the AA feature has a positive impact, suggesting that during the pandemic, airlines prioritized maintaining strong, well-established connections rather than creating new ones.

Overall, the analysis illustrates the pandemic-driven shift toward regional optimization and decreased reliance on hub-centric networks. Although the core hubs remained influential, their functionality weakened, necessitating a focus on regional routes and efficient resource allocation to maintain network operability during a period of unprecedented uncertainty.

(3) The Post-Pandemic Period (2023–2024):

The results in Fig. 8 for the post-pandemic period show a shift in the Chinese aviation network from regionalization back to decentralization, while simultaneously emphasizing the importance of network redundancy and diversity. The PA feature remains the top feature in both models, with positive SHAP values showing that core hub airports have regained their ability to attract route connections. Similarly, the HP feature is much more important in both models in the post-pandemic period than in the pandemic period, reflecting the crucial role of core hubs in restoring network connectivity. The RA feature ranks second in the unweighted model, showing a notable increase in importance compared with that during the pandemic period. This suggests that airlines placed greater emphasis on optimizing the allocation of limited resources to routes and airports with the highest operational and economic value. While the importance of LHN has slightly decreased in the post-pandemic phase—likely due to the recovery of larger hub airports—its continued prominence indicates that regional connectivity remains a crucial component of the aviation network, supporting distributed network resilience and reducing dependency on a few major hubs. With the tightening of Chinese airspace and slot regulations during the pandemic recovery period (CAAC, 2022), major hub-based airlines benefited from policies that favor their operations, accelerating the development of multiple hubs. In contrast, the enthusiasm for point-to-point routes connecting smaller and secondary cities decreased, reflecting airlines' prioritization of long-haul and high-demand routes while maintaining key regional connections. Additionally, the importance of the JA and SA features increases in the unweighted model. This

increase suggests a growing focus on network redundancy and diversity in the post-pandemic recovery phase, enhancing its capacity to adapt to potential future risks and disruptions. Consistent with Chen and Chen (2025)'s emphasis on enhancing system resilience against external disruptions, our findings highlight the necessity of a robust strategy to maintain operational stability under the pandemic and resource constraints.

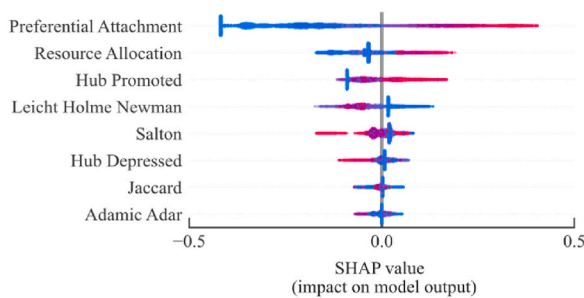
These trends collectively demonstrate that although the recovery process is marked by a return to centralization around major hubs, there is also a renewed emphasis on balancing efficiency with resilience through enhanced connectivity among regional nodes and the promotion of network diversity.

4.4. Identification of important features

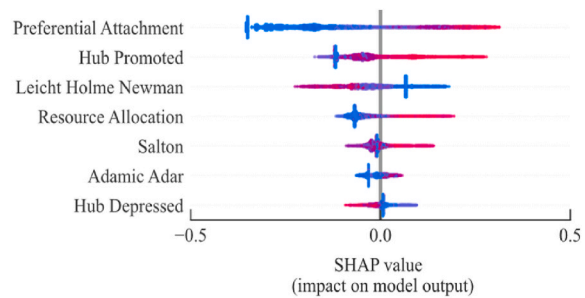
The feature importance rankings in Fig. 9(a)–(c) show that the monthly importance of aviation network topological features from 2014 to 2024 reflects the structural adjustments of the Chinese aviation network across the pre-pandemic, pandemic, and post-pandemic periods. The heatmap's color gradient, ranging from light to dark, indicates the importance of each feature, with darker bars signifying greater importance in the corresponding snapshot. Fig. 9 highlights that the LHN, HP, RA, and PA features are the four key factors in predicting annual network dynamics.

The PA feature consistently ranks as the most important feature between 2014 and 2019, underscoring the long-term and stable development of the aviation network's centralization. In the pre-pandemic period, the PA feature drives the centralization of the network, with hub airports continuously consolidating their dominance through the addition of new connections. However, airlines preferred to maintain existing routes and critical connections during the pandemic, leading to a decreased reliance on adding new routes.

The RA feature maintains a consistently high ranking during 2014–2019, reflecting airlines' strategies to enhance network efficiency by optimizing route allocation and airport resource utilization. Although the RA feature's importance shows minor fluctuations, it remains at a consistently high level. As shown in Fig. 9(b), the importance of the RA feature drops by two positions during the pandemic because airlines prioritized core routes and key regional links to minimize the impact of route disruptions on network connectivity. This indicates the critical role of the RA feature in mitigating the adverse effects of reduced capacity during this period. In the post-pandemic recovery period, the importance of the RA feature greatly increases, reflecting airlines' increased reliance on efficient resource allocation to meet the rapidly growing demand as the network recovered. Furthermore, the LHN feature shows an increase in importance followed by a decline. The increased importance of the LHN feature means that airlines enhanced connectivity among smaller airports, improving regional coverage and redundancy to better respond to shifting market demands and



(a) Unweighted topological features.



(b) Weighted topological features.

Fig. 8. (a) Unweighted topological features (b) Weighted topological features SHAP values of two models between 2023 and 2024.

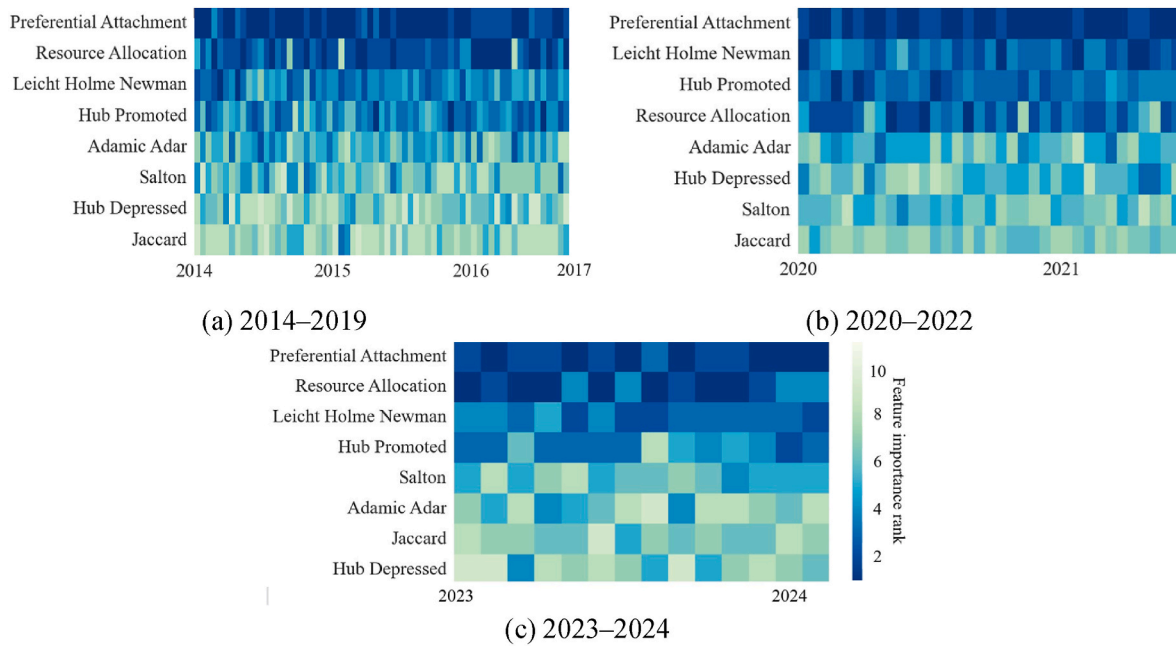


Fig. 9. (a) 2014–2019 (b) 2020–2022 (c) 2023–2024 Feature importance rankings for snapshots of 2014–2024 according to their SHAP values.

strengthen network adaptability. Additionally, the HD feature follows a similar pattern, with its importance ranking increasing initially and then decreasing. Before the pandemic, this feature has a relatively small negative impact, mainly reflecting the loss of connectivity for hub airports due to market competition or resource constraints. Lastly, the importance of the JA and SA features is relatively low from 2014 to 2019. However, their importance increases substantially in the post-pandemic phase, indicating a strategic shift to improving network diversity and redundancy as airlines strengthen secondary hubs and regional connections to enhance the flexibility and resilience of the network.

Overall, these results highlight the dynamic evolution of the importance of topological features across different periods, reflecting the adaptive strategies of airlines in response to changing operational challenges and market demands. The emphasis on efficient resource allocation, regional connectivity, and network redundancy underscores the importance of balancing efficiency, resilience, and adaptability in

aviation network planning and recovery.

5. Discussion

While the events examined in this study span the pre-pandemic, pandemic, and post-pandemic periods, the structural changes observed during each phase offer valuable insights for future aviation planning. The following discussion draws on empirical findings to highlight implications that may inform future responses to large-scale disruptions or long-term shifts in network demand and design priorities.

(1) The pre-pandemic period (2014–2019):

Fig. 10 presents snapshots of the Chinese aviation network in October 2016 and October 2019. The red circles representing airports indicate the number of outbound flight routes from each airport, with the size of the circle corresponding to the number of routes. The blue

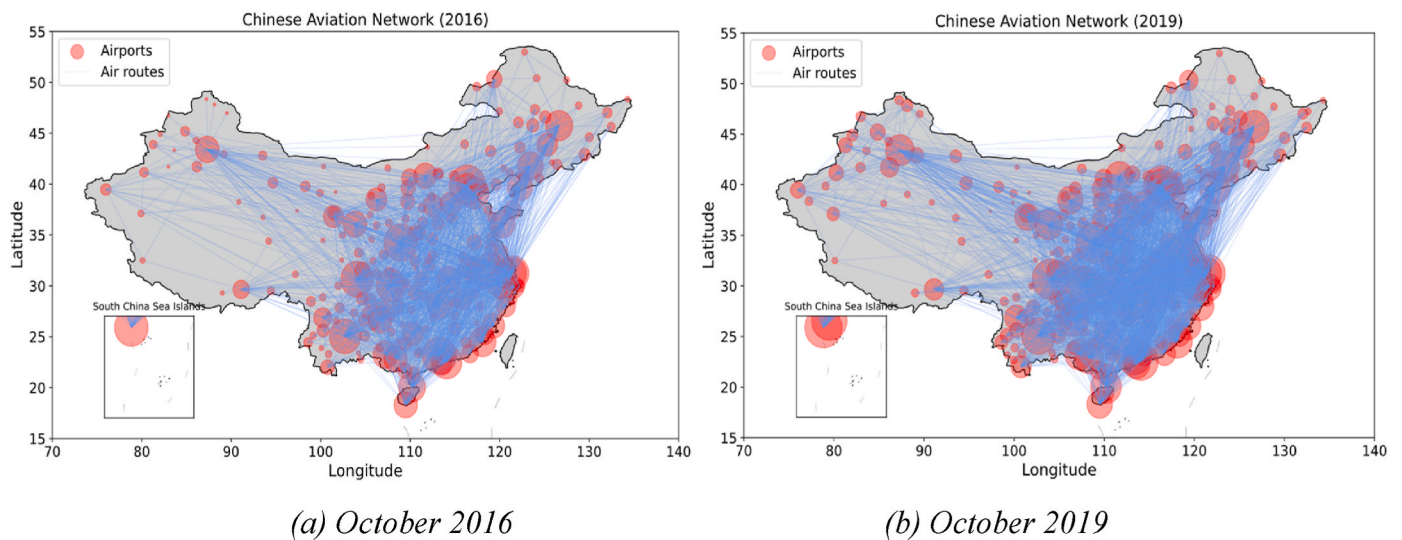


Fig. 10. (a) October 2016 (b) October 2019 Snapshots of the Chinese aviation network in October 2016 and October 2019.

lines represent flight routes connecting the various airports within the network.

Before the COVID-19 pandemic, the evolution of the Chinese aviation network was shaped by a combination of market-driven expansion and state-led development. Policies such as the 13th Five-Year Plan for Civil Aviation (2016–2020) encouraged large-scale construction at major hubs (Xinhua News, 2021). Projects such as the completion of Guangzhou Baiyun Terminal 2 in 2018 (Wan et al., 2024) and the preparatory works for Beijing Daxing International Airport (Beijing Municipal Commission of Development and Reform, 2021) reflected a national emphasis on building global aviation hubs to consolidate Chinese international and domestic air connectivity.

In this policy environment, airlines pursued strategies that concentrated routes and resources around core airports to maximize scale economies and reduce operating costs. As shown in Fig. 10, this led to increasing centralization of the network between 2016 and 2019. Topological indicators such as PA and RA features rose during this period, reflecting a strong tendency for new connections to attach to already well-connected nodes. The formation of hub-and-spoke configurations became dominant, with large carriers using central hubs to aggregate traffic flows and improve aircraft utilization rates. While this configuration improved network efficiency and enabled rapid traffic growth in the late 2010s, it also introduced structural fragility. The over-concentration of traffic at a few key nodes, such as Beijing Capital, increased the risk of systemic disruptions. For instance, a single adverse event at one of these hubs (e.g., heavy fog, slot saturation, or security lockdown) had the potential to create cascading delays throughout the network due to the limited redundancy built into the system.

These observations lead to several key network design insights:

1. Core hubs serve as foundational nodes that enable route development and support the spatial coverage of airline networks (Yu et al., 2024). Sustained investment in hub airport capacity remains necessary to ensure that dominant nodes can accommodate increasing volumes.
2. Developing alternative inter-regional routes can help maintain operations during future disruptions (Y. Zhang et al., 2024). Specifically, fostering the development of secondary hubs and regional airports can increase redundancy and mitigate centralization risk. This approach is consistent with the "One Core, Two Wings, Multiple Nodes" framework introduced by the CAAC to promote multi-level hub development.

3. To avoid monopolization of prime airport infrastructure, competitive slot allocation mechanisms are essential. Evidence from cities like Shenzhen and Chengdu, where multiple carriers operate from shared terminals, shows that balanced slot distribution can enhance competition and reduce systemic vulnerability (Tran et al., 2017).

(2) The pandemic period (2020–2022):

During the COVID-19 pandemic, the Chinese aviation network underwent an abrupt decentralization, reflecting shifts in regulatory conditions and operational strategy. In early 2020, international and interprovincial travel restrictions were implemented under China's "Containment policy", which severely reduced long-haul and transnational flights ((Civil Aviation Administration of China, 2020a)). Airlines responded by concentrating on short-haul domestic routes and reducing overall network complexity to contain operational risks and cost. This transition was facilitated by government-led "rescue policies", including temporary slot relaxation at regional airports ((Civil Aviation Administration of China, 2020b)) and direct subsidies for domestic and regional flights (Hou et al., 2021), especially in the western provinces and third-tier cities.

As Fig. 11 shows, topological indicators from this period show a decline in centrality and average degree, accompanied by a growing role of LHN and AA features, which capture localized and similarity-driven link formation. During this period, airlines focused on maintaining existing connections and essential regional linkages, with a reduced role for the PA feature, which aligns with the suspension of hub expansion plans during the pandemic. This structural simplification helped sustain baseline connectivity but also exposed the vulnerability of regional airports and secondary cities, many of which faced drastic drops in load factors and profitability. In response, the CAAC introduced special regional connectivity funds, such as subsidies to support routes to Tibet, Xinjiang, and other underserved areas, ensuring the operational viability of these critical nodes during the crisis.

These dynamics lead to the following policy insights:

1. Government subsidies for short-haul and feeder routes play a critical role in sustaining connectivity and preserving access for geographically vulnerable areas during the pandemic.
2. Stabilizing hub airports through targeted financial assistance is also vital during the pandemic. Research shows that disruptions at major hubs produce disproportionate impacts across the

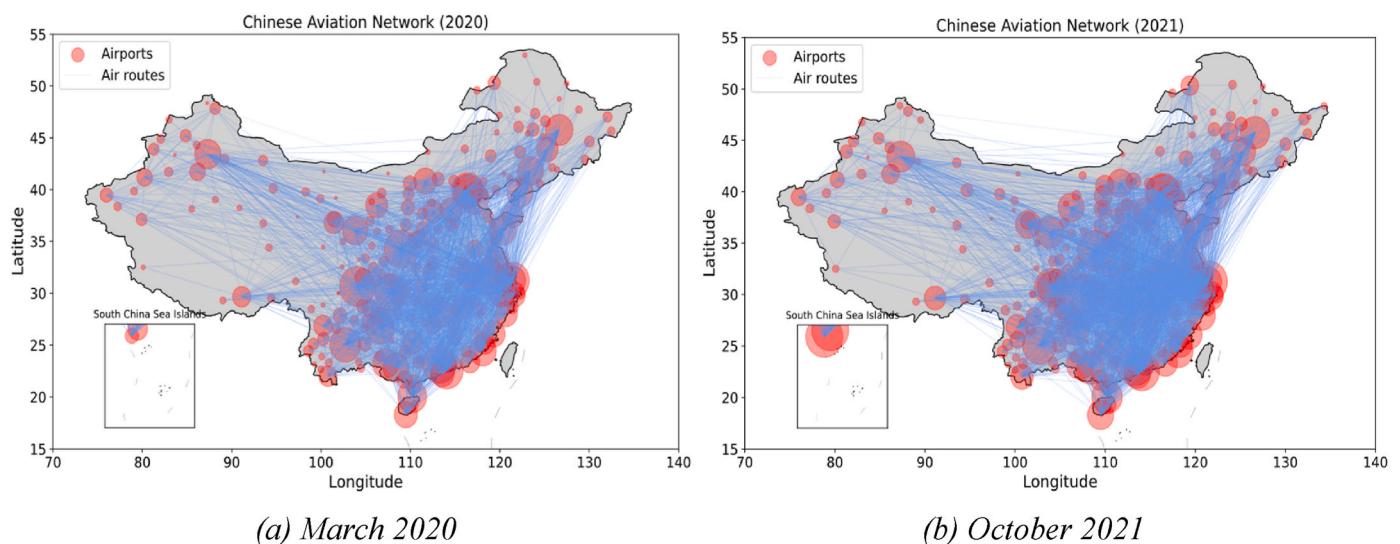


Fig. 11. (a) March 2020 (b) October 2021
Snapshots of the Chinese aviation network in March 2020 and October 2021.

network (Bombelli et al., 2020). Maintaining strategic core routes can preserve a large share of international connectivity even under global travel restrictions (Li et al., 2021b).

- Promoting operational flexibility and regulatory responsiveness can enhance aviation system resilience. Airlines should be allowed to dynamically adjust schedules and capacities in response to rapidly evolving demand conditions. This aligns with evidence from Zhu et al. (2021) and Li et al. (2024), which shows that flexible policy environments enable more effective recovery and resource reallocation.

(3) The post-pandemic period (2023–2024):

Following the containment of the COVID-19 pandemic, the Chinese civil aviation network entered a gradual recovery phase characterized by the return of passenger demand. This resurgence was shaped by national recovery policies, such as the CAAC's "Five-One" rollback policy (Civil Aviation Administration of China, 2022) and provincial-level incentives aimed at restoring intercity and cross-border air links. Major airlines resumed route expansion plans, with a renewed emphasis on trunk routes and hub airports, supported by funding for terminal expansion at airports like Shanghai Pudong and Guangzhou Baiyun (Civil Aviation Administration of China, 2024).

Topologically, this recovery is reflected in the recentralization of the network (Fig. 12), with the RA feature regaining prominence. Airlines increasingly concentrated capacity and scheduling around dominant hubs to capture resurgent demand, reflecting a return to efficiency-oriented network design. However, the declining influence of the HP feature suggests that although primary hubs recovered first, the overall restoration of connectivity remained asymmetric and was conducted across different tiers of airports. For instance, while Tier 1 hubs saw rapid recovery of international routes by mid-2023, many regional airports continued to operate below pre-pandemic capacity. Moreover, the reduced importance of the JA and SA features reflects a shift in priority away from network diversity and redundancy. This pattern suggests that airlines chose to concentrate their operations on a smaller group of well-performing routes, rather than restoring lower-demand connections. It highlights a strategy that emphasizes resilience through the stability of core routes rather than through widespread redundancy across the network.

These patterns suggest two key post-pandemic policy insights:

- Targeted investment in hub airport recovery is essential. Governments should continue supporting infrastructure upgrades and provide financial incentives to restore long-haul and international routes at major hubs. Recent subsidies for direct flights between China and Europe via Beijing and Shanghai have played a key role in reviving high-value traffic and bilateral connectivity.
- Balancing network centralization with resilience remains a key challenge. While hub-focused strategies improve network efficiency, overreliance on a few major airports can recreate the vulnerabilities exposed during the pandemic. A coordinated multi-hub model, such as the combined use of Beijing Daxing and Capital airports, helps reduce congestion and improve redundancy (Calzada and Fageda, 2023). Policies that support multi-hub development, such as performance-based subsidies, can enhance spatial equity, expand service coverage, and strengthen overall network robustness (Cheung et al., 2020; Li et al., 2023).

(4) Cross-period reflections and strategic directions

The evolving patterns observed across pre-pandemic, pandemic, and post-pandemic periods reveal how external shocks and policy interventions collectively shape the structure of aviation networks. These shifts underscore the need for future aviation systems to strike a dynamic balance between efficiency, resilience, and adaptability.

In the pre-pandemic period, airlines adopted centralized hub-and-spoke networks to achieve economies of scale, improve load factors, and reduce operating costs. Government infrastructure investments reinforced this trend by concentrating resources in a few major airports. However, such centralization created systemic vulnerabilities, increasing the risk of widespread disruption when key hubs were affected.

The COVID-19 pandemic exposed these weaknesses. Mobility restrictions and route limitations quickly disrupted demand and route viability. In response, airlines shifted toward short-haul and regional services, leading to more decentralized network structures. This shift underscored the importance of regional connectivity in maintaining basic mobility and access during crises.

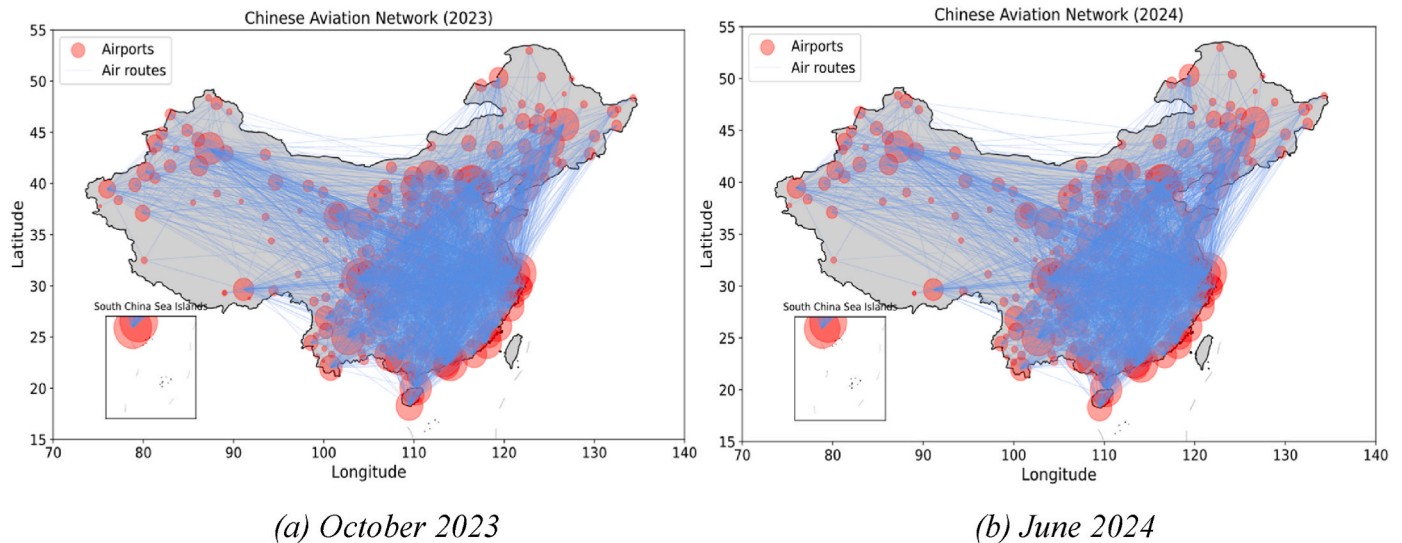


Fig. 12. (a) October 2023 (b) June 2024
Snapshots of the Chinese aviation network in October 2023 and June 2024.

In the post-pandemic recovery phase, recentralization has resumed as airlines focus on restoring efficiency and network connectivity. However, the uneven recovery across airport tiers and the selective return of long-haul routes reflect persistent structural imbalances and evolving market conditions. These trends highlight the ongoing challenge of balancing efficiency with resilience.

In summary, the structural evolution of the Chinese aviation network across these three distinct periods reflects the ongoing tension between optimization and resilience. By grounding network design in both empirical insights and contextual understanding, policymakers can guide the aviation sector toward a future that is not only more efficient, but also better prepared to withstand uncertainty and disruption.

6. Conclusions

This study analyzes the evolution of the Chinese aviation network across the pre-pandemic, pandemic, and post-pandemic periods. We propose an ML-based framework that integrates link prediction methods to indicate the network's evolutionary dynamics and calculate the contributions of 11 key topological features that drive these changes. By incorporating actual OD passenger flow data, we also reveal how the importance of critical topological features shifted over time in response to the disruptions caused by the COVID-19 pandemic.

Based on the SHAP analysis and identification of the important features driving the evolution of the Chinese aviation network, several policy insights emerge. First, the dominance of the PA feature highlights the critical role of core hub airports in maintaining network connectivity, particularly during the recovery phase. Policymakers need to provide targeted investment in major hubs such as Beijing, Shanghai, and Guangzhou to meet the post-pandemic demand and reinforce their roles as central nodes in the aviation network. Second, the increasing importance of the RA feature underscores the need for efficient resource management strategies. Airlines and regulators optimize resource allocation by prioritizing routes and airports that enhance network stability and passenger demand, particularly during periods of disruption and recovery. Furthermore, the role of the LHN feature indicates the continued importance of regional and domestic connectivity, particularly when international routes are disrupted. Policies that encourage the development of secondary hubs and enhance regional routes would improve network redundancy and resilience. Finally, the increasing importance of diversity-related features such as the JA and SA features after the pandemic suggests a shift toward prioritizing network robustness and flexibility. Policymakers should incentivize airlines to diversify their route networks, reducing overreliance on a few major hubs and promoting alternative connections that enhance adaptability to future disruptions. By systematically examining shifting preferences in aviation network design (centralization, resource optimization, and regional connectivity), this study provides a forward-looking perspective on

sustainable and adaptive network planning.

This study is subject to the following limitations. First, the analysis primarily focuses on the aviation network and does not account for the competition and interplay between air travel and other modes of transportation such as high-speed rail. Well-developed high-speed rail greatly influences passenger demand and network connectivity, which could alter the evolution of the aviation network. Future studies could incorporate multimodal transportation networks to provide a more comprehensive understanding of these dynamics. Second, although this study evaluates the structural evolution of the aviation network, it does not explicitly consider the strategic decisions of, and competition among, airlines. Airline-specific factors, such as route planning strategies, pricing mechanisms, and alliances, can significantly influence network dynamics. Incorporating airline competition and behavior into the analysis would provide a more nuanced perspective on network evolution.

CRedit authorship contribution statement

Ziyu Cui: Writing – original draft, Validation, Methodology, Formal analysis, Data curation, Conceptualization. **Hanjiang Dong:** Writing – original draft, Validation, Methodology, Formal analysis, Data curation, Conceptualization. **Kun Wang:** Writing – original draft, Methodology, Investigation, Formal analysis, Conceptualization. **Jiehong Qiu:** Writing – original draft, Methodology, Investigation, Data curation. **Xiaowen Fu:** Writing – original draft, Supervision, Resources, Project administration, Methodology, Conceptualization.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used ChatGPT 4.0 to grammar check the initial draft. After using this tool, the authors reviewed and edited the manuscript as needed and took full responsibility for the content of the publication.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have influenced the work reported in this paper.

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Appendix A

A1. Summary of input variables in ML-based model

Table A.6

Summary of input variables in ML-based model.

Type	Input variables	Example
Network characteristics	Link	('nkg', 'jgd')
Time	Year	January 2023 (month/year)
Similarity scores of features	Common Neighbor	8
	Salton	0.265
	Jaccard	0.078
	Sorensen	0.145
	Hub Promoted	0.889
	Hub Depressed	0.079
	Leicht-Holme-Newman	0.009
	Preferential Attachment	909
	Adamic-Adar	1.755
	Resource Allocation	0.089
	Local Path	0
	Weighted Common Neighbor	11
	Weighted Salton	0.197
	Weighted Jaccard	0.042
	Weighted Sorensen	0.081
	Weighted Hub Promoted	0.917
	Weighted Hub Depressed	0.042
	Weighted Leicht-Holme-Newman	0
	Weighted Preferential Attachment	3108
	Weighted Adamic-Adar	16.469
	Weighted Resource Allocation	0.475
	Weighted Local Path	0
Decision	Whether the link is added this month compared to last month	TURE/FALSE

A2. ML-based algorithm pseudocode

Algorithm 1 Machine learning algorithms based on link prediction

1: Input:

- 2: Potential link set for future network G_{t+1} , similarity-based index scores, addition labels, machine learning models (e.g., XGBoost and Random Forest), top L links to select.
- 3: **Output:** Predicted link set $\hat{E}_{t+1}$.
- 4: **Step 1: Generate potential links**
- 5: **for** each node pair $(i, j) \in V_t$
- 6: Define potential links in the network G_{t+1} .
- 7: **Step 2: Calculate similarity-based scores and assign labels**
- 8: **for** $\forall (i, j) \in V_t$ **do**
- 9: Calculate similarity-based index scores (e.g., Resource Allocation, Preferential Attachment).
- 10: Assign addition labels for each potential link (i, j) based on similarity scores.
- 11: **Step 3: Predict link existence probabilities**
- 12: **for** each potential link with a given similarity score **do**
- 13: Use machine learning models to predict the probability of existence for link (i, j) .
- 14: **Step 4: Rank candidate links**
- 15: Sort all candidate links (i, j) by their predicted probabilities in descending order.
- 16: **Step 5: Select top L Links**
- 17: **for** top L ranked links in sorted list **do**
- 18: Add (i, j) to the predicted link set $\hat{E}_{t+1}$.
- 19: **Step 6: Evaluate predictive performance**
- 20: Compare $\hat{E}_{t+1}$ with ground truth links in G_{t+1} to assess predictive performance of the model.
- 21: **Return** $\hat{E}_{t+1}$

Data availability

The authors do not have permission to share data.

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